The intestinal microflora in allergic Estonian and Swedish 2-year-old children.

BACKGROUND: The prevalence of allergic diseases seems to have increased particularly over the past 35-40 years. Furthermore, allergic disease is less common among children in the formerly socialist countries of central and Eastern Europe as compared with Western Europe. It has been suggested that a reduced microbial stimulation during infancy and early childhood would result in a slower postnatal maturation of the immune system and development of an optimal balance between TH1- and TH2-like immunity. AIMS: To test the hypothesis that allergic disease among children may be associated with differences in their intestinal microflora in two countries with a low (Estonia) and a high (Sweden) prevalence of allergy. METHODS: From a prospective study of the development of allergy in relation to environmental factors, 29 Estonian and 33 Swedish 2-year-old children were selected. They were either nonallergic ($n = 36$) or had a confirmed diagnosis of allergy ($n = 27$) as verified by typical history and at least one positive skin prick test to egg or cow's milk. Weighed samples of faeces were serially diluted (10-2-10-9) and grown under anaerobic conditions. The counts of the various genera and species were calculated for each child. In addition, the relative amounts of the particular microbes were expressed as a proportion of the total count. RESULTS: The allergic children in Estonia and Sweden were less often colonized with lactobacilli ($P < 0.01$), as compared with the nonallergic children in the two countries. In contrast, the allergic children harboured higher counts of aerobic micro-organisms ($P < 0. 05$), particularly coliforms ($P < 0.01$) and Staphylococcus aureus ($P < 0.05$). The proportions of aerobic bacteria of the intestinal flora were also higher in the allergic children ($P < 0.05$), while the opposite was true for anaerobes ($P < 0.05$). Similarly, in the allergic children the proportions of coliforms were higher ($P < 0. 05$) and bacteroides lower ($P < 0.05$) than in the nonallergic children. CONCLUSIONS: Differences in the indigenous intestinal flora might affect the development and priming of the immune system in early childhood, similar to what has been shown in rodents. The role of intestinal microflora in relation to the development of infant immunity and the possible consequences for allergic diseases later in life requires further study, particularly as it would be readily available for intervention as a means for primary prevention of allergy by the administration of probiotic bacteria.